

Zhuo (Prince) Niu

UI/UX Designer & AI Product Developer | Human Factors Engineering + Computer Engineering

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SUMMARY

“Bridging Code & Design for Scalable Web and Mobile Experiences”

- Human Factors Engineering M.S. student with a Computer Engineering background, focused on human-AI interaction.
- Build end-to-end mobile and macOS products, from interaction design to production-minded systems.
- Translate user needs into structured interactions, high-fidelity interfaces, and working software.

EDUCATION

Tufts University Medford, MA • Sep 2025 – Expected Dec 2026
Master of Science, Human Factors Engineering
Relevant Courses: Human-Machine System Design, Computer Interface Design, Advanced Research Methods, Applied Behavioral Statistics, Human Factors Product Design, Use-Related Risk Management

Purdue University West Lafayette, IN • Aug 2020 – Dec 2024
Bachelor of Science, Computer Engineering
Relevant Courses: Advanced C Programming, Data Structures, Microprocessor Systems and Interfacing, Computer Graphics, Operating Systems, Data Science, Artificial Intelligence, Computer Security, Data Mining

WORK EXPERIENCE

Porsche (Porsche Asia Pacific) Hong Kong, China SAR
UI/UX Designer Jun 2024 – Sep 2024

- Collaborated with **cross-functional design and engineering teams** to prototype UI components for Porsche’s in-car system, supporting both touch and voice-based navigation.
- Developed **high-fidelity Figma prototypes and component specs** to clarify interaction details and streamline frontend handoff.
- Presented **design rationale and interaction flows** to align team members on usability goals, visual consistency, and technical feasibility.
- Optimized UI responsiveness and micro-interactions through iterative feedback, contributing to a **30% reduction in load time**.

China GPI Tech Shanghai, China
User System Developer Jun 2023 – Aug 2023

- Collaborated with **product and engineering teams** to design multimodal UX flows integrating voice, touch, and GUI interactions for an AI-driven user system.
- Mapped **user interaction scenarios** and refined NLP voice commands, improving speech recognition accuracy to **95%**.
- Presented and documented **input transition logic** to align developers on smoother handoff between voice and touch commands.

TECHNICAL SKILLS

AI & Agentic Tools
Prompt Engineering, AI-Assisted Coding, OpenAI Codex, Claude Code, RAG Systems, Dify, AI Agent Workflows, Human-AI Interaction, Whisper / Speech-to-Text

Programming & App Development
Python, Swift, JavaScript, TypeScript, React, React Native, HTML, CSS, Tailwind CSS, C/C++

Tools, APIs & Data
Git, GitHub, Firebase, Expo, Xcode, APNs, ActivityKit, WidgetKit, LocalAuthentication, SSE, RESTful APIs, GraphQL, SQL, PostgreSQL, FastAPI

UI & Interaction Design
Figma, Design Systems, UI Components, Material Design, Human Interface Guidelines, Human-Centered Design, User Research, Usability Testing, Typography, Color Theory

PROJECTS

Hermes iOS Companion

Jul 2026 – Present

- Built a **SwiftUI mobile companion for the self-hosted Hermes Agent**, supporting session continuity, streamed tool execution, voice input, and isolated Mac and VPS profiles over Tailscale.
- Implemented **SSE reconnect and state reconciliation**, an offline Share outbox, instance-specific APNs routing, scoped approvals, and device authentication for app access and Gateway restart.
- Validated the system with **595 unit tests, 2 focused text-selection UI tests**, and repeated physical-iPhone acceptance covering APNs, Share Extension, Widget, Live Activity, bilingual UI, and accessibility.
- **Tools Used:** Swift, SwiftUI, SSE, APNs, ActivityKit, WidgetKit, LocalAuthentication, Speech, Share Extension, Tailscale, Xcode

KnowledgeOS

May 2026

- Built a **personal knowledge system** where an AI agent handles classification, deduplication, and review across **100+ Markdown files, 400+ wikilinks, and 0 broken links**.
- Defined rules for what gets saved, how it is categorized, and when duplicates merge, then automated daily capture, weekly synthesis, and maintenance audits.
- Released an **open-source Starter Vault** with setup guides, example workflows, customization documentation, and validation scripts.
- **Tools Used:** Python, Markdown, Obsidian, Git, GitHub, Hermes Agent, Automation, Agent Workflow Design

SayIt

Mar 2026 – Apr 2026

- Developed a **production-ready macOS menu bar app** for voice-to-text transcription with global hotkey recording, live status feedback, and automatic clipboard copy.
- Integrated Apple Speech and on-device Whisper through **whisper.cpp**, supporting privacy-focused transcription with configurable engines and local model support.
- Built a polished **SwiftUI/AppKit interface** with microphone selection, audio-level visualization, HUD notifications, and bilingual localization.
- Shipped a **downloadable v1.0.5 release** through GitHub Releases with DMG packaging, Developer ID signing, notarization, Sparkle updates, and a versioned release workflow.
- **Tools Used:** Swift, SwiftUI, AppKit, AVFoundation, Speech Framework, CoreAudio, whisper.cpp, Xcode, GitHub

TrekAssist

Sep 2025 – Dec 2025

- Collaborated on a **Figma prototype** for a solar-powered e-ink national park guide designed for Yellowstone visitors under low-power, **1 Hz refresh-rate display constraints**.
- Conducted **usability testing with 9 participants**, tracking task success, completion time, errors, attempts, and ease ratings across six outdoor-use scenarios.
- Translated test findings into **design updates**, improving alert differentiation, battery-setting access, map readability, and persistent emergency access.
- Designed **safety-centered interaction flows** for weather warnings, ranger SOS confirmation, trail navigation, guided tours, visitor safety tips, and low-power settings.
- **Tools Used:** Figma, FigJam, Usability Testing, User Flows, Prototyping, Excel

PROFESSIONAL CERTIFICATES

Google UX Design Professional Certificate

2024

Coursera

- **Skills Acquired:** User Research, Wireframing, Prototyping, Figma, User Experience Design, Usability, Mockups, Low-Fidelity and High-Fidelity Prototypes, User Interface

Meta iOS Developer Professional Certificate

2024

Coursera

- **Skills Acquired:** Swift, Apple iOS, React Native, iOS Development, Core Data, React.js, Version Control, UI/UX Research, Design and Product